Test 2

Assignment

- G signifies students that take this class for **Graduate Credit**; they must do all the items
- UG students are **not** required to do the G work, but they may do it for bonus points
- Problems solved beyond the minimum requirements will be given bonus points.
- Challenge problems are optional; they will earn bonus points.

NOTES:

- Show your work to obtain partial credit if appropriate
- Always display input data!
- Partial credit might be given based on work done and MATLAB codes if they are attached; without them, no partial credit can be given.
- Display numerical results to three significant digits (four if the first digit is 1).
- Use SI system of measurement https://en.wikipedia.org/wiki/International_System_of_Units
- Scale units using appropriate prefixes (e.g., nano, micro, milli, kilo, mega, giga…) https://en.wikipedia.org/wiki/Metric_prefix
- Problems solved beyond the minimum requirements will be given bonus points.
- Challenge problems are optional. They may be solved for extra credit
- Normalized plots are plots with the *y*-axis scaled between –1 and +1. A plotted variable is normalized by dividing it by its maximum absolute value

Notations and abbreviations:

- BVP = boundary value problem
- EOM = equation of motion
- FBD = free body diagram
- FRF = frequency response function
- IVP = initial value problem
- LHS = left hand side
- NLM = Newton's law of motion
- ODE = ordinary differential equation
- PDE = partial differential equation
- RHS = right hand side

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A. Waves

A.1 WAVE FUNDAMENTALS

NOTHING HERE

A.2 SIMPLE WAVE CALCULATIONS

NOTHING HERE

A.3 CALCULATE CAR MOTION ON WAVY ROAD

Consider a car traveling on a wavy road. The road surface undulates with an amplitude a=6-in and a wavelength λ=140 ft that follows a cosine shape. The car is modeled as a spring-mass-damper (Figure 1) with natural frequency f_n=0.96 Hz and damping ratio ζ=14%. The maximum car speed is v_{max}=113 mph.

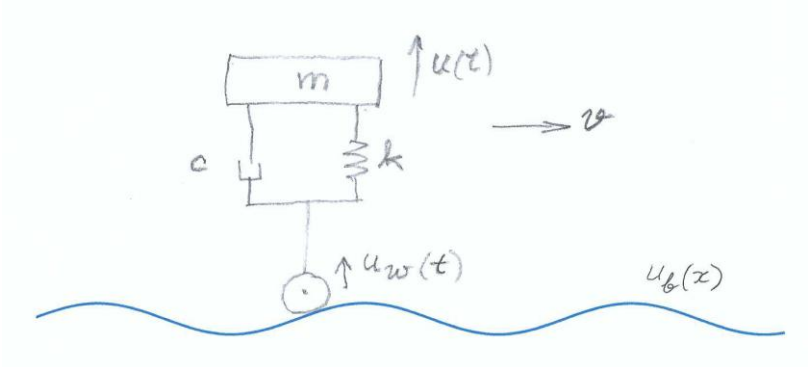


Figure 1 Car riding on a wavy road

Do the following:

(1) Display input data
(2) Calculate wavy road wavenumber γ
(3) Plot the wavy road profile $u_b(x)$ for N_λ=2 wavelengths
(4) calculate wavy-road frequency f_{max} for a maximum car speed of v_{max} and define the frequency range f=linspace(0,f_{max},N_f) with N_f=2000
(5) plot $|FRF(f\,;f_n)|$ for. Use datatips on the plot to locate the peak response and the matched-frequency response.
(6) Calculate the matched-frequency response value $|FRF_n|$. Compare with the matched-frequency readings at item (5)
(7) Calculate the car speed v_n needed to make wavy-road frequency f match the car natural frequency f_n

(8) Use the conversion $f=v/\lambda$ to plot $|FRF(v\ ;f_n)|$. Use datatips on the plot to find:
   (a) v_n and $|FRF_n|$ and discuss them in comparison with results of items (6), (7)
   (b) v_{cr} and $|FRF_{max}|$
   (c) the limit speed v_{limit} for which the motion amplification is limited to $|FRF_{lim}|=1.5$

(9) **Graduate work:** Find the road wavelength λ_{new} that will make v_{cr} =30 mph

(10) **Graduate work:** Repeat items (5) through (8) for the case of a car with worn-out shocks that can only give ζ=10% and comment on the comparison between the two cases

A.4 STANDING WAVES

(1) nothing here
(2) nothing here
(3) Assume wavespeed c=340m/s and frequency f=3.6kHz. Calculate:
   (a) rad/sec frequency ω
   (b) wavenumber γ
   (c) wavelength λ
   (d) distance Δx between standing wave nodes
(4) nothing here

A.5 MUSICAL NOTES

(1) Find the Hz frequency of note E4
(2) Find what musical note corresponds to 1396.9 Hz
(3) What is the commonly accepted frequency range of human hearing?
(4) What is the lowest frequency on the scale of equal temperament and what musical note does it correspond to?
(5) What is the highest frequency on the scale of equal temperament and what musical note does it correspond to?
(6) **Graduate Work**: explain the scale of equal temperament:
    (a) what is the organizing principle of the scale of equal temperament
    (b) how many equal-temperament entries are in an octave?
    (c) what is the ratio between two adjacent entries on the scale of equal temperament?

A.6 MUSICAL INSTRUMENTS

Nothing here

B. Fluid Waves and Vibration

B.1 ANALYZE WAVE EQUATION IN FLUIDS

Nothing here

B.2 ANALYZE AND FIND SOLUTION OF WAVE EQUATION

Nothing here

B.3 ANALYZE AIR VIBRATION IN A TUBE CLOSED AT BOTH ENDS

Nothing here

B.4 CALCULATE AIR VIBRATION IN A TUBE CLOSED AT BOTH ENDS

Consider a tube of length L=0.5 m closed at both ends. Take sound speed c=340 m/s and air density ρ=1.2 kg/m^3. Use the results of Problem B.3 to calculate the modal vibration characteristics and the modeshapes for j=1,…,N modes with N=5.

(1) Display input data
(2) Calculate and display the modal vibration characteristics:
    (a) eigenvalues z_j
    (b) wavenumbers γ_j
    (c) rad/s frequencies ω_j
    (d) Hz frequencies f_j
(3) Calculate and plot the displacement modeshapes
(4) Discuss your results
(5) **Graduate work**: Calculate and plot the normalized pressure modeshapes and discuss your result in comparison with item (3)

B.5 CALCULATE THE SLIDE WHISTLE

Consider the slide whistle shown in Figure 2. Assume sound speed in air to be c=340m/s. The length of the residual cavity of the fully retracted whistle tube can be approximated to be L_0=10 mm.

- Take the length of the fully extended whistle as L_e=420 mm.
- Take the length of the fully contracted whistle as L_c=260 mm.

The purpose of this problem is to calculate the fundamental frequency for various positions of the whistle slider

(a)



(b)



Figure 2 sliding whistle: (a) fully retracted, showing the tip of the piston and the residual cavity (b) fully extended.

Do the following:

(1) display input data c and L_0
(2) display the fully extended whistle length L_e
(3) display the fully contracted whistle length L_c
(4) use L_e, L_c to calculate the extension travel dL_I corresponding to fully extended slide whistle
(5) use dL_I and L_0 to calculate the cavity length L_I of a fully extended slide whistle
(6) estimate the fully extended frequency f_I in Hz and find the nearest musical note expressed as a letter name and an octave number (e.g., G3 signifies note G in the 3rd octave)
(7) use L_e, L_c to calculate the extension travel dL_{II} corresponding to 50% extended slide whistle
(8) use dL_{II} and L_0 to calculate the cavity length L_{II} of a half-extended slide whistle

(9) calculate the half-extended frequency f_{II} in Hz and find the nearest musical note expressed as a letter name and an octave number

(10) comment on your work

Graduate work: Estimate the position of the piston to obtain the musical note D7. Do the following:

(11) display the frequency f_{III} of the note D7

(12) calculate the cavity length L_{III} needed to generate frequency f_{III} of the note D7

(13) calculate the extension travel dL_{III} needed to generate frequency f_{III} of the note D7

(14) estimate what percentage of the piston has to be extended to generate the note D7

(15) comment on your work

B.6 CALCULATE SOUND INTENSITY, dB SCALE

Take sound speed c=340 m/s and air density ρ=1.2 kg/m^3. Do the following:

(1) Display input data
(2) Calculate air's acoustic impedance Z
(3) Calculate the sound intensity I corresponding to a sound pressure p=88 mPa
(4) Give the values of the reference sound intensity I_0 and reference sound pressure p_0
(5) Covert to dB the sound pressure and the sound intensity of item (3) above
(6) Verify that the sound pressure and sound intensity calculated at item (5) have the same dB value and explain why it is so
(7) Calculate in physical units the sound intensity, sound pressure, and sound power from a loudspeaker with diameter D=1.25 m working at 72 dB
(8) Calculate the sound pressure p_{horn} created by a 124 dB car horn
(9) Give the values of the pain sound intensity I_{pain} and pain sound pressure p_{pain} and comment on the results of items (7) and (8) relative to pain values
(10) **Graduate Work:** explain why it is necessary to use different dB conversion formulae for sound intensity I and sound pressure p.

C. Extra Credit Challenge Problems

Challenge problems are optional.

C.1 ANALYZE AIR VIBRATION IN A CLOSED-OPEN TUBE

NOTHING HERE

C.2 ANALYZE AIR VIBRATION IN AN OPEN-OPEN TUBE

NOTHING HERE

C.3 CALCULATE AIR VIBRATION IN A TUBE WITH VARIOUS BOUNDARY CONDITIONS

Consider a tube of length L=0.5 m. Take sound speed c=340 m/s and air density ρ=1.2 kg/m^3. The purpose is to calculate the modal vibration characteristics and the modeshapes for j=1,…,N modes with N=5. The following boundary conditions will be considered:

boundary conditions		
case	x=0	x=L
(i)	closed	open
(ii)	open	open

(1) Display input data

Then, for each of the two boundary conditions cases, do the following:

(2) Calculate and display the modal vibration characteristics:
   (a) eigenvalues z_j
   (b) wavenumbers γ_j
   (c) rad/s frequencies ω_j
   (d) Hz frequencies f_j

(3) Calculate and plot the displacement modeshapes

(4) Discuss your results in comparison with Problem B.4

(5) **Graduate work**: Calculate and plot the normalized pressure modeshapes and discuss your result in comparison with item (3)